# 23. PWM Delay Generation Circuit (PDG)

## 23.1 Overview

The PWM Delay Generation circuit (PDG) has 4 channels delay circuits that can connect to the GPT. The PDG can control the rise and fall edge timing with which the PWM output for the GPT320 through the GPT323.

Table 23.1 lists the specifications for the PWM Delay Generation Circuit, Figure 23.1 shows a block diagram, and Table 23.2 lists the I/O pins.

**Table 23.1 Specifications of the PWM Delay Generation Circuit**

| Parameter | Specifications |
|---|---|
| Function | The circuit can control the timing with which signals on the two PWM output pins for channel 0/1/2/3 rise and fall to an accuracy of up to 1/128 times the period of the GPT core clock (GTCLK).<br>The GPT core clock (GTCLK) can be selected from PCLKD or GPTCLK. |

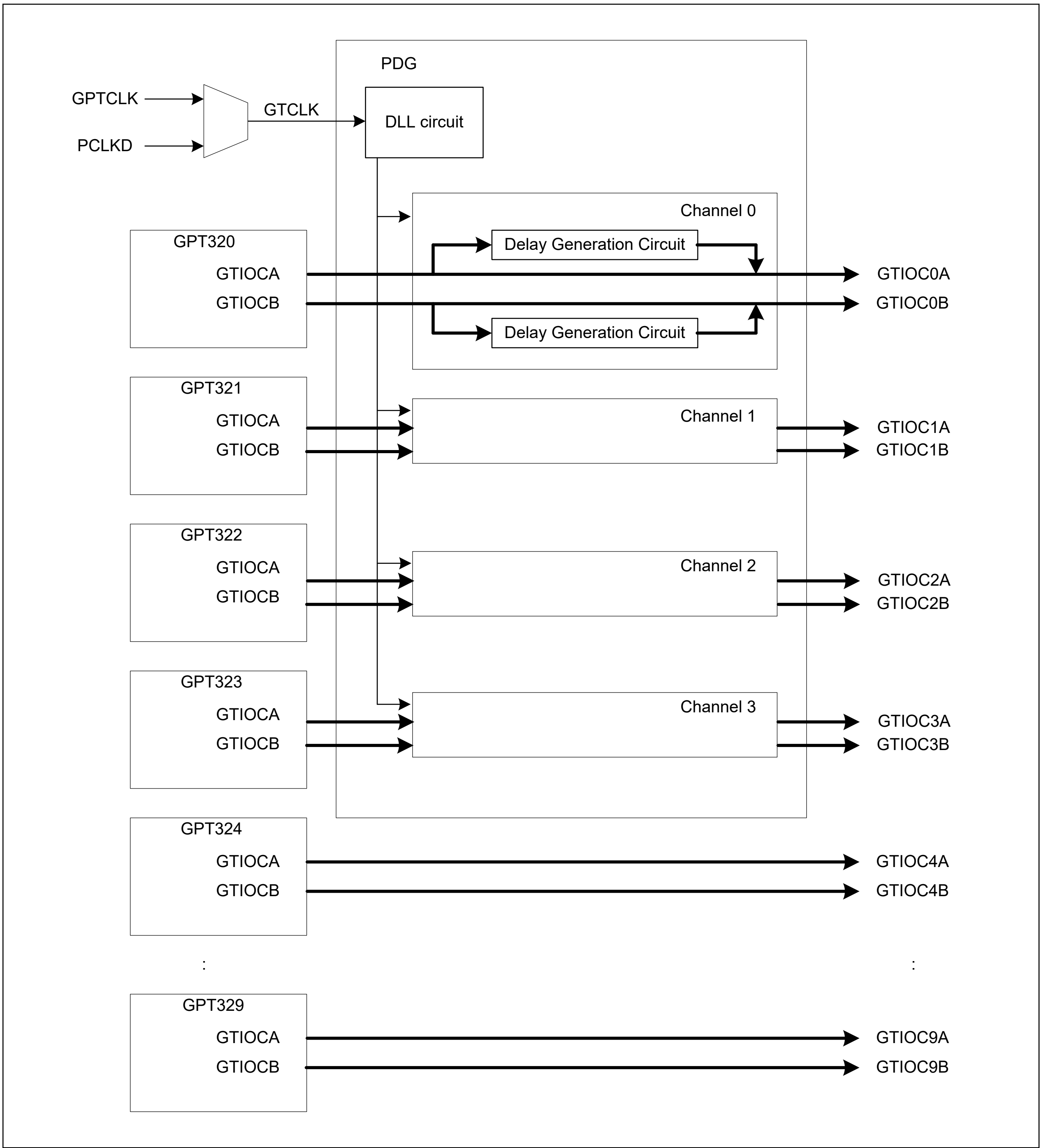



**Figure 23.1 PWM delay generation circuit block diagram**

**Table 23.2 PWM delay generation circuit I/O pins (1 of 2)**

| I/O pin | I/O | Function |
|---|---|---|
| GTIOC0A | Output | Delayed output of GTIOCA pin of GPT channel 0 |
| GTIOC0B | Output | Delayed output of GTIOCB pin of GPT channel 0 |
| GTIOC1A | Output | Delayed output of GTIOCA pin of GPT channel 1 |
| GTIOC1B | Output | Delayed output of GTIOCB pin of GPT channel 1 |
| GTIOC2A | Output | Delayed output of GTIOCA pin of GPT channel 2 |

**Table 23.2 PWM delay generation circuit I/O pins (2 of 2)**

| I/O pin | I/O | Function |
|---|---|---|
| GTIOC2B | Output | Delayed output of GTIOCB pin of GPT channel 2 |
| GTIOC3A | Output | Delayed output of GTIOCA pin of GPT channel 3 |
| GTIOC3B | Output | Delayed output of GTIOCB pin of GPT channel 3 |

## 23.2 Register Descriptions

### 23.2.1 GTDLYCR : PWM Output Delay Control Register

Base address: PDG = 0x4032_4000
PDG_NS = 0x5032_4000

Offset address: 0x0000

| Bit position: | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Bit field: | — | — | — | — | — | — | FRANGE[1:0] | | — | — | — | — | — | — | DLYRST | DLLEN |
| Value after reset: | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |

| Bit | Symbol | Function | R/W |
|---|---|---|---|
| 0 | DLLEN | DLL Operation Enable<br>0: DLL operation disabled<br>1: DLL operation enabled | R/W |
| 1 | DLYRST | PWM Delay Generation Circuit Reset<br>0: Normal operation<br>1: Reset | R/W |
| 7:2 | — | These bits are read as 0. The write value should be 0. | R/W |
| 9:8 | FRANGE[1:0] | GPT core clock Frequency Range<br>0 0: GPT core clock frequency is 80 MHz to 160 MHz<br>0 1 GPT core clock frequency is 155 MHz to 300 MHz<br>Others: Setting prohibited | R/W |
| 15:10 | — | These bits are read as 0. The write value should be 0. | R/W |

The GTDLYCR register controls the PWM delay generation circuit, which applies delays to the PWM outputs. GTDLYCR register can be written when register write protection is disabled (GPT320.GTWP.WP = 0).

When changing GTDLYCR after changing the value of GPT320.GTWP.WP bit, be sure to read back the value of GTWP register before changing the value of GTDLYCR.

#### DLLEN bit (DLL Operation Enable)

The DLLEN bit selects whether the on-chip DLL in the PWM delay generation circuit is activated or not.

#### DLYRST bit (PWM Delay Generation Circuit Reset)

The DLYRST bit resets the internal state of the PWM delay generation circuit.

#### FRANGE[1:0] bit (GPT core clock Frequency Range)

The FRANGE[1:0] bits set the frequency range of the GPT core clock.

Set the FRANGE[1:0] bits only when the DLLEN bit is 0.

## 23.2.2 GTDLYCR2 : PWM Output Delay Control Register 2

Base address: PDG = 0x4032_4000
PDG_NS = 0x5032_4000

Offset address: 0x0002

| Bit position: | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Bit field: | — | — | — | — | DLYEN3 | DLYEN2 | DLYEN1 | DLYEN0 | — | — | — | — | DLYBS3 | DLYBS2 | DLYBS1 | DLYBS0 |
| Value after reset: | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |

| Bit | Symbol | Function | R/W |
|---|---|---|---|
| 0 | DLYBS0 | PWM Delay Generation Circuit bypass for channel 0<br>0: Delay generation circuit of channel 0 bypassed<br>1: Delay generation circuit of channel 0 not bypassed | R/W |
| 1 | DLYBS1 | PWM Delay Generation Circuit bypass for channel 1<br>0: Delay generation circuit of channel 1 bypassed<br>1: Delay generation circuit of channel 1 not bypassed | R/W |
| 2 | DLYBS2 | PWM Delay Generation Circuit bypass for channel 2<br>0: Delay generation circuit of channel 2 bypassed<br>1: Delay generation circuit of channel 2 not bypassed | R/W |
| 3 | DLYBS3 | PWM Delay Generation Circuit bypass for channel 3<br>0: Delay generation circuit of channel 3 bypassed<br>1: Delay generation circuit of channel 3 not bypassed | R/W |
| 7:4 | — | These bits are read as 0. The write value should be 0. | R/W |
| 8 | DLYEN0 | PWM Delay Generation Circuit enable for channel 0<br>0: Delay generation circuit of channel 0 enabled<br>1: Delay generation circuit of channel 0 disabled | R/W |
| 9 | DLYEN1 | PWM Delay Generation Circuit enable for channel 1<br>0: Delay generation circuit of channel 1 enabled<br>1: Delay generation circuit of channel 1 disabled | R/W |
| 10 | DLYEN2 | PWM Delay Generation Circuit enable for channel 2<br>0: Delay generation circuit of channel 2 enabled<br>1: Delay generation circuit of channel 2 disabled | R/W |
| 11 | DLYEN3 | PWM Delay Generation Circuit enable for channel 3<br>0: Delay generation circuit of channel 3 enabled<br>1: Delay generation circuit of channel 3 disabled | R/W |
| 15:12 | — | These bits are read as 0. The write value should be 0. | R/W |

The GTDLYCR2 register controls each channel of PWM delay generation circuit. GTDLYCR2 can be written when register write protection is disabled (GPT320.GTWP.WP = 0).

When changing GTDLYCR2 after changing the value of GPT320.GTWP.WP bit, be sure to read back the value of GTWP register before changing the value of GTDLYCR2.

### DLYBSn (n = 0 to 3) bit (PWM Delay Generation Circuit Bypass for channel n)

The DLYBSn bit selects whether delays are applied to PWM output signals from the GTIOCnA and GTIOCnB pins (n = 0 to 3) by the PWM delay generation circuit or whether the circuit is bypassed.

A signal delayed in the PWM delay generation circuit is output 3 cycles of GPT core clock (GTCLK) later than if it bypasses the PWM delay generation circuit.

### DLYENn (n = 0 to 3) bit (PWM Delay Generation Circuit Enable for channel n)

The DLYENn bit selects whether channel n (n = 0 to 3) of PWM delay generation circuit is power on or off. If channel n of the PWM delay generation circuit is not used, set to 1.

### 23.2.3 GTDLYRnA : GTIOCnA Rising Output Delay Register (n = 0 to 3)

Base address: PDG = 0x4032_4000
PDG_NS = 0x5032_4000

Offset address: 0x018 + 0x4 × n

| Bit position: | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Bit field: | — | — | — | — | — | — | — | — | — | DLY[6:0] | | | | | | |
| Value after reset: | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |

| Bit | Symbol | Function | R/W |
|---|---|---|---|
| 6:0 | DLY[6:0] | GTIOCnA Output Rising Edge Delay Setting<br>(1) Case of GTDLYCR.FRANGE[1:0] = 00b<br>0x00: Delay on rising edges is not applied.<br>0x01: Delay of 1/128 times GTCLK period applied.<br>0x02: Delay of 2/128 times GTCLK period applied.<br>0x03: Delay of 3/128 times GTCLK period applied.<br>0x04: Delay of 4/128 times GTCLK period applied.<br>0x05: Delay of 5/128 times GTCLK period applied.<br>0x06: Delay of 6/128 times GTCLK period applied.<br>⋮<br>0x7B: Delay of 123/128 times GTCLK period applied.<br>0x7C: Delay of 124/128 times GTCLK period applied.<br>0x7D: Delay of 125/128 times GTCLK period applied.<br>0x7E: Delay of 126/128 times GTCLK period applied.<br>0x7F: Delay of 127/128 times GTCLK period applied.<br>(2) Case of GTDLYCR.FRANGE[1:0] = 01b<br>0x00: Delay on rising edges is not applied.<br>0x01: Delay on rising edges is not applied.<br>0x02: Delay of 1/64 times GTCLK period applied.<br>0x03: Delay of 1/64 times GTCLK period applied.<br>0x04: Delay of 2/64 times GTCLK period applied.<br>0x05: Delay of 2/64 times GTCLK period applied.<br>0x06: Delay of 3/64 times GTCLK period applied.<br>⋮<br>0x7B: Delay of 61/64 times GTCLK period applied.<br>0x7C: Delay of 62/64 times GTCLK period applied.<br>0x7D: Delay of 62/64 times GTCLK period applied.<br>0x7E: Delay of 63/64 times GTCLK period applied.<br>0x7F: Delay of 63/64 times GTCLK period applied. | R/W |
| 15:7 | — | These bits are read as 0. The write value should be 0. | R/W |

The GTDLYRnA register sets a delay to be applied to rising edges of output signals on the GTIOCnA pin. On the timing for the transfer of settings, see section 23.3.2. Timing for Transfer of GTDLYRnA, GTDLYRnB, GTDLYFnA, and GTDLYFnB Register Settings.

GTDLYRnA can be written when register write protection is disabled (GPT32n.GTWP.WP = 0).

When changing GTDLYRnA after changing the value of GPT32n.GTWP.WP bit, be sure to read back the value of GTWP register before changing the value of GTDLYRnA.

### 23.2.4 GTDLYFnA : GTIOCnA Falling Output Delay Register (n = 0 to 3)

Base address: PDG = 0x4032_4000
PDG_NS = 0x5032_4000

Offset address: 0x028 + 0x4 × n

| Bit position: | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Bit field: | — | — | — | — | — | — | — | — | — | DLY[6:0] | | | | | | |
| Value after reset: | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |

| Bit | Symbol | Function | R/W |
|---|---|---|---|
| 6:0 | DLY[6:0] | GTIOCnA Output Falling Edge Delay Setting<br>(1) Case of GTDLYCR.FRANGE[1:0] = 00b<br>0x00: Delay on rising edges is not applied.<br>0x01: Delay of 1/128 times GTCLK period applied.<br>0x02: Delay of 2/128 times GTCLK period applied.<br>0x03: Delay of 3/128 times GTCLK period applied.<br>0x04: Delay of 4/128 times GTCLK period applied.<br>0x05: Delay of 5/128 times GTCLK period applied.<br>0x06: Delay of 6/128 times GTCLK period applied.<br>⋮<br>0x7B: Delay of 123/128 times GTCLK period applied.<br>0x7C: Delay of 124/128 times GTCLK period applied.<br>0x7D: Delay of 125/128 times GTCLK period applied.<br>0x7E: Delay of 126/128 times GTCLK period applied.<br>0x7F: Delay of 127/128 times GTCLK period applied.<br>(2) Case of GTDLYCR.FRANGE[1:0] = 01b<br>0x00: Delay on rising edges is not applied.<br>0x01: Delay on rising edges is not applied.<br>0x02: Delay of 1/64 times GTCLK period applied.<br>0x03: Delay of 1/64 times GTCLK period applied.<br>0x04: Delay of 2/64 times GTCLK period applied.<br>0x05: Delay of 2/64 times GTCLK period applied.<br>0x06: Delay of 3/64 times GTCLK period applied.<br>⋮<br>0x7B: Delay of 61/64 times GTCLK period applied.<br>0x7C: Delay of 62/64 times GTCLK period applied.<br>0x7D: Delay of 62/64 times GTCLK period applied.<br>0x7E: Delay of 63/64 times GTCLK period applied.<br>0x7F: Delay of 63/64 times GTCLK period applied. | R/W |
| 15:7 | — | These bits are read as 0. The write value should be 0. | R/W |

The GTDLYFnA register sets a delay to be applied to falling edges of output signals on the GTIOCnA pin. On the timing for the transfer of settings, see section 23.3.2. Timing for Transfer of GTDLYRnA, GTDLYRnB, GTDLYFnA, and GTDLYFnB Register Settings.

GTDLYFnA can be written when register write protection is disabled (GPT32n.GTWP.WP = 0).

When changing GTDLYFnA after changing the value of GPT32n.GTWP.WP bit, be sure to read back the value of GTWP register before changing the value of GTDLYFnA.

## 23.2.5 GTDLYRnB : GTIOCnB Rising Output Delay Register (n = 0 to 3)

Base address: PDG = 0x4032_4000
PDG_NS = 0x5032_4000

Offset address: 0x01A + 0x4 × n

| Bit position: | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Bit field: | — | — | — | — | — | — | — | — | — | DLY[6:0] | | | | | | |
| Value after reset: | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |

| Bit | Symbol | Function | R/W |
|---|---|---|---|
| 6:0 | DLY[6:0] | GTIOCnB Output Rising Edge Delay Setting<br>(1) Case of GTDLYCR.FRANGE[1:0] = 00b<br>0x00: Delay on rising edges is not applied.<br>0x01: Delay of 1/128 times GTCLK period applied.<br>0x02: Delay of 2/128 times GTCLK period applied.<br>0x03: Delay of 3/128 times GTCLK period applied.<br>0x04: Delay of 4/128 times GTCLK period applied.<br>0x05: Delay of 5/128 times GTCLK period applied.<br>0x06: Delay of 6/128 times GTCLK period applied.<br>⋮<br>0x7B: Delay of 123/128 times GTCLK period applied.<br>0x7C: Delay of 124/128 times GTCLK period applied.<br>0x7D: Delay of 125/128 times GTCLK period applied.<br>0x7E: Delay of 126/128 times GTCLK period applied.<br>0x7F: Delay of 127/128 times GTCLK period applied.<br>(2) Case of GTDLYCR.FRANGE[1:0] = 01b<br>0x00: Delay on rising edges is not applied.<br>0x01: Delay on rising edges is not applied.<br>0x02: Delay of 1/64 times GTCLK period applied.<br>0x03: Delay of 1/64 times GTCLK period applied.<br>0x04: Delay of 2/64 times GTCLK period applied.<br>0x05: Delay of 2/64 times GTCLK period applied.<br>0x06: Delay of 3/64 times GTCLK period applied.<br>⋮<br>0x7B: Delay of 61/64 times GTCLK period applied.<br>0x7C: Delay of 62/64 times GTCLK period applied.<br>0x7D: Delay of 62/64 times GTCLK period applied.<br>0x7E: Delay of 63/64 times GTCLK period applied.<br>0x7F: Delay of 63/64 times GTCLK period applied. | R/W |
| 15:7 | — | These bits are read as 0. The write value should be 0. | R/W |

The GTDLYRnB register sets a delay to be applied to rising edges of output signals on the GTIOCnB pin. On the timing for the transfer of settings, see section 23.3.2. Timing for Transfer of GTDLYRnA, GTDLYRnB, GTDLYFnA, and GTDLYFnB Register Settings.

GTDLYRnB can be written when register write protection is disabled (GPT32n.GTWP.WP = 0).

When changing GTDLYRnB after changing the value of GPT32n.GTWP.WP bit, be sure to read back the value of GTWP register before changing the value of GTDLYRnB.

## 23.2.6 GTDLYFnB : GTIOCnB Falling Output Delay Register (n = 0 to 3)

Base address: PDG = 0x4032_4000
PDG_NS = 0x5032_4000

Offset address: 0x02A + 0x4 × n

| Bit position: | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Bit field: | — | — | — | — | — | — | — | — | — | DLY[6:0] | | | | | | |
| Value after reset: | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |

| Bit | Symbol | Function | R/W |
|---|---|---|---|
| 6:0 | DLY[6:0] | GTIOCnB Output Falling Edge Delay Setting<br>(1) Case of GTDLYCR.FRANGE[1:0] = 00b<br>0x00: Delay on rising edges is not applied.<br>0x01: Delay of 1/128 times GTCLK period applied.<br>0x02: Delay of 2/128 times GTCLK period applied.<br>0x03: Delay of 3/128 times GTCLK period applied.<br>0x04: Delay of 4/128 times GTCLK period applied.<br>0x05: Delay of 5/128 times GTCLK period applied.<br>0x06: Delay of 6/128 times GTCLK period applied.<br>⋮<br>0x7B: Delay of 123/128 times GTCLK period applied.<br>0x7C: Delay of 124/128 times GTCLK period applied.<br>0x7D: Delay of 125/128 times GTCLK period applied.<br>0x7E: Delay of 126/128 times GTCLK period applied.<br>0x7F: Delay of 127/128 times GTCLK period applied.<br>(2) Case of GTDLYCR.FRANGE[1:0] = 01b<br>0x00: Delay on rising edges is not applied.<br>0x01: Delay on rising edges is not applied.<br>0x02: Delay of 1/64 times GTCLK period applied.<br>0x03: Delay of 1/64 times GTCLK period applied.<br>0x04: Delay of 2/64 times GTCLK period applied.<br>0x05: Delay of 2/64 times GTCLK period applied.<br>0x06: Delay of 3/64 times GTCLK period applied.<br>⋮<br>0x7B: Delay of 61/64 times GTCLK period applied.<br>0x7C: Delay of 62/64 times GTCLK period applied.<br>0x7D: Delay of 62/64 times GTCLK period applied.<br>0x7E: Delay of 63/64 times GTCLK period applied.<br>0x7F: Delay of 63/64 times GTCLK period applied. | R/W |
| 15:7 | — | These bits are read as 0. The write value should be 0. | R/W |

The GTDLYFnB register sets a delay to be applied to falling edges of output signals on the GTIOCnB pin. On the timing for the transfer of settings, see section 23.3.2. Timing for Transfer of GTDLYRnA, GTDLYRnB, GTDLYFnA, and GTDLYFnB Register Settings.

GTDLYFnB can be written when register write protection is disabled (GPT32n.GTWP.WP = 0).

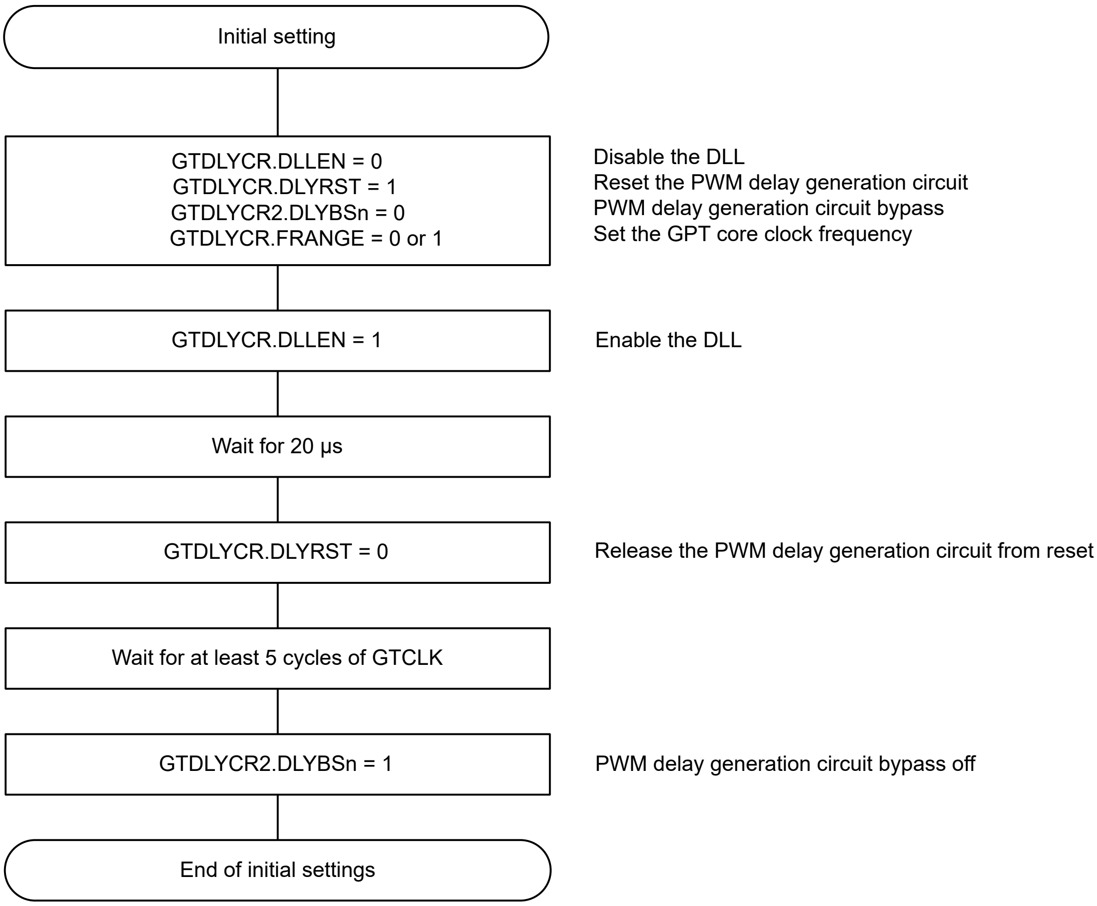
When changing GTDLYFnB after changing the value of GPT32n.GTWP.WP bit, be sure to read back the value of GTWP register before changing the value of GTDLYFnB.

## 23.3 Operation

### 23.3.1 Adjustments to the Timing of Rising and Falling Edges in PWM Waveforms

The timing of rising and falling edges in PWM waveforms which are output from the GTIOCnA and GTIOCnB pins, where n = channel number, can be delayed to an accuracy of 1/64 or 1/128 of the GPT core clock (GTCLK) period.

If the timing of rising or falling edges in PWM waveforms output from the GTIOCnA and GTIOCnB pins must be adjusted, initialize the PWM generation circuit as shown in the procedure in Figure 23.2.




Note: n = 0 to 3

**Figure 23.2 Example of initialization flow for the PWM delay generation circuit**

In the PWM delay generation circuit, delay can be applied to rising and falling edges of the PWM output to an accuracy of 1/64 or 1/128 of the period of the GPT core clock (GTCLK). This is described in section 22.3.3. PWM Output Operating Mode. Delays associated with the settings are reflected in the PWM output with the timing described in section 23.3.2. Timing for Transfer of GTDLYRnA, GTDLYRnB, GTDLYFnA, and GTDLYFnB Register Settings. Table 23.3 shows the association between the GTDLYRnA, GTDLYRnB, GTDLYFnA, and GTDLYFnB registers and the PWM outputs.

**Table 23.3 Association between PWM output pins and delay setting registers**

| PWM output pin | Rising-edge delay setting register | Falling-edge delay setting register |
|---|---|---|
| GTIOC0A | GTDLYR0A | GTDLYF0A |
| GTIOC0B | GTDLYR0B | GTDLYF0B |
| GTIOC1A | GTDLYR1A | GTDLYF1A |
| GTIOC1B | GTDLYR1B | GTDLYF1B |
| GTIOC2A | GTDLYR2A | GTDLYF2A |
| GTIOC2B | GTDLYR2B | GTDLYF2B |
| GTIOC3A | GTDLYR3A | GTDLYF3A |
| GTIOC3B | GTDLYR3B | GTDLYF3B |

When the PWM delay generation circuit is in use, the timing with which a PWM output signal rises and falls can be controlled to an accuracy of 1/64 or 1/128 of the period of the GPT core clock (GTCLK). When this option is not in use, the period of the PWM output waveform is controlled to an accuracy of one period of the input clock for the timer counter, which is GTCLK. With the PWM delay generation circuit, the output can be controlled to an accuracy 32 times better.

Additionally, the delay settings also control the periods at high and low level for the PWM waveform to the given accuracy. PWM delay generation circuit channels can be individually enabled or disabled.

### 23.3.2 Timing for Transfer of GTDLYRnA, GTDLYRnB, GTDLYFnA, and GTDLYFnB Register Settings

Settings for the GTDLYRnA, GTDLYRnB, GTDLYFnA, and GTDLYFnB registers are initially transferred to temporary registers, and then reflected in the delay on the GTIOCnA and GTIOCnB (n = 0 to 3) outputs. Transfer of the settings takes place on overflows (in up-counting) or underflows (in down-counting) for saw waves, and in the troughs of triangle waves.

Figure 23.3 and Figure 23.4 show examples of the operation of the GTDLYR0A and GTDLYF0A registers.

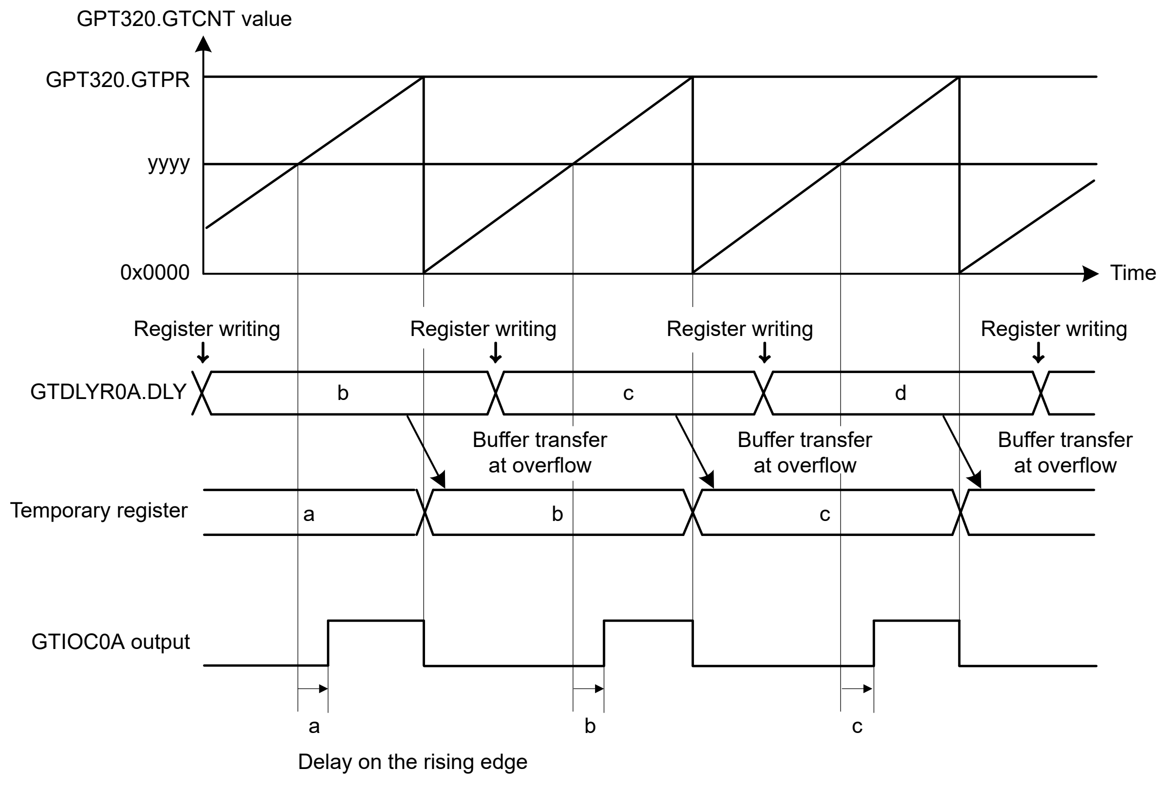



**Figure 23.3 Example of GTDLYR0A register operation with PWM saw-wave generation**

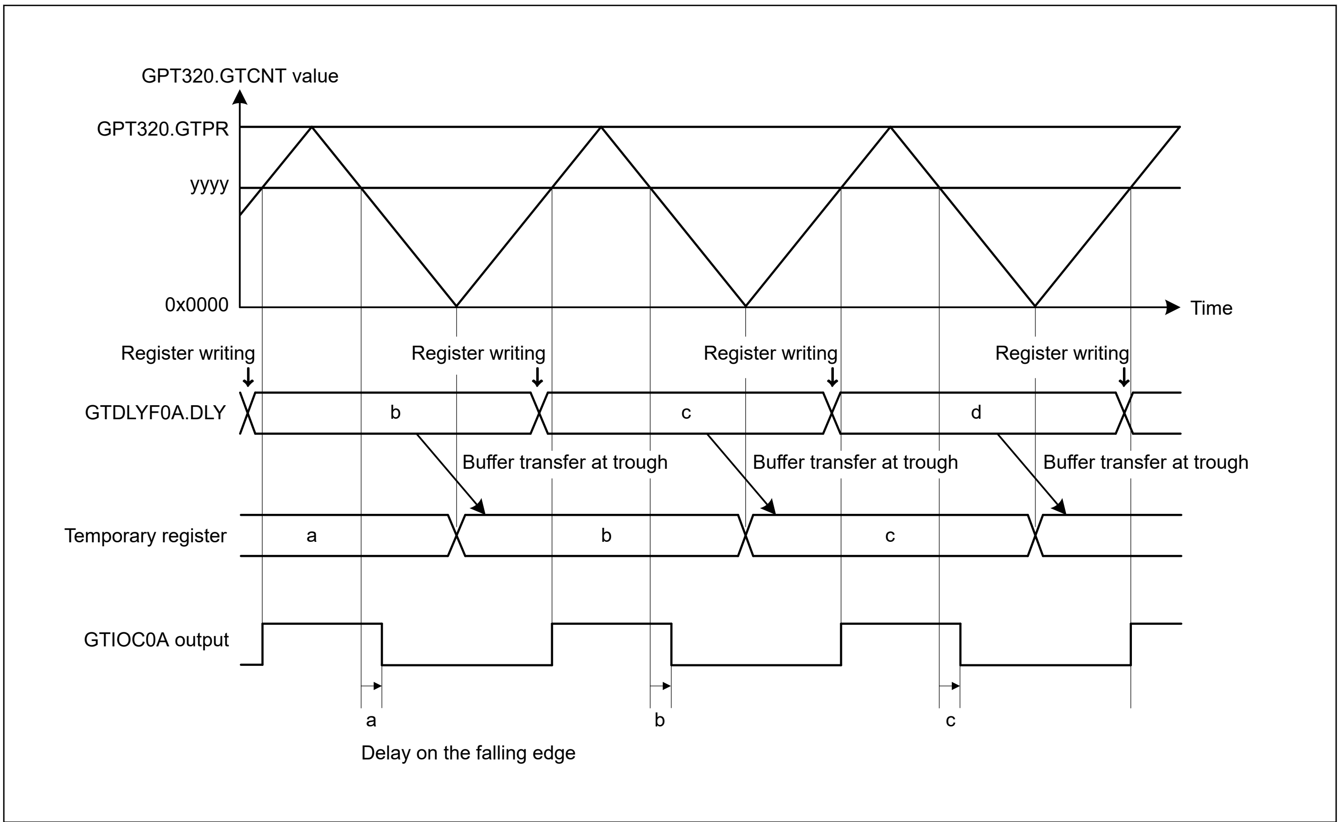



**Figure 23.4 Example of GTDLYF0A register operation with PWM triangle-wave generation**

## 23.4 Usage Notes

### 23.4.1 Settings for the Module-Stop Function

The Module Stop Control Register D (MSTPCRD) can enable or disable operation of the PWM delay generation circuit. The PWM delay generation circuit is initially stopped after reset. Releasing the module-stop state enables access to the registers. For details, see section 11, Low Power Mode.

### 23.4.2 Notes on Delay Settings for PWM Delay Generation Circuit

When the PWM delay generation circuit generates delays for a PWM output waveform and the waveform is toggled in response to compare-matches, do not change the settings for delay while the compare-match value is within the ranges listed in Table 23.4. This constraint applies to the GTDLYFnA, GTDLYRnA, GTDLYFnB, and GTDLYRnB registers.

**Table 23.4 Constraints on delay settings**

| Mode | Direction of counting | Compare-match value |
|---|---|---|
| Saw-wave mode | Up | GTPR - 2 or above |
| | Down | 2 or below |
| Triangle-wave mode | Down | 2 or below |

Figure 23.5 shows an example of how the constraints apply to the timing of setting GTDLYFnA in saw-wave waveform one-shot pulse mode (counting up). Do not change the value set in GTDLYFnA while GTCCRD ≥ GTPR - 2.

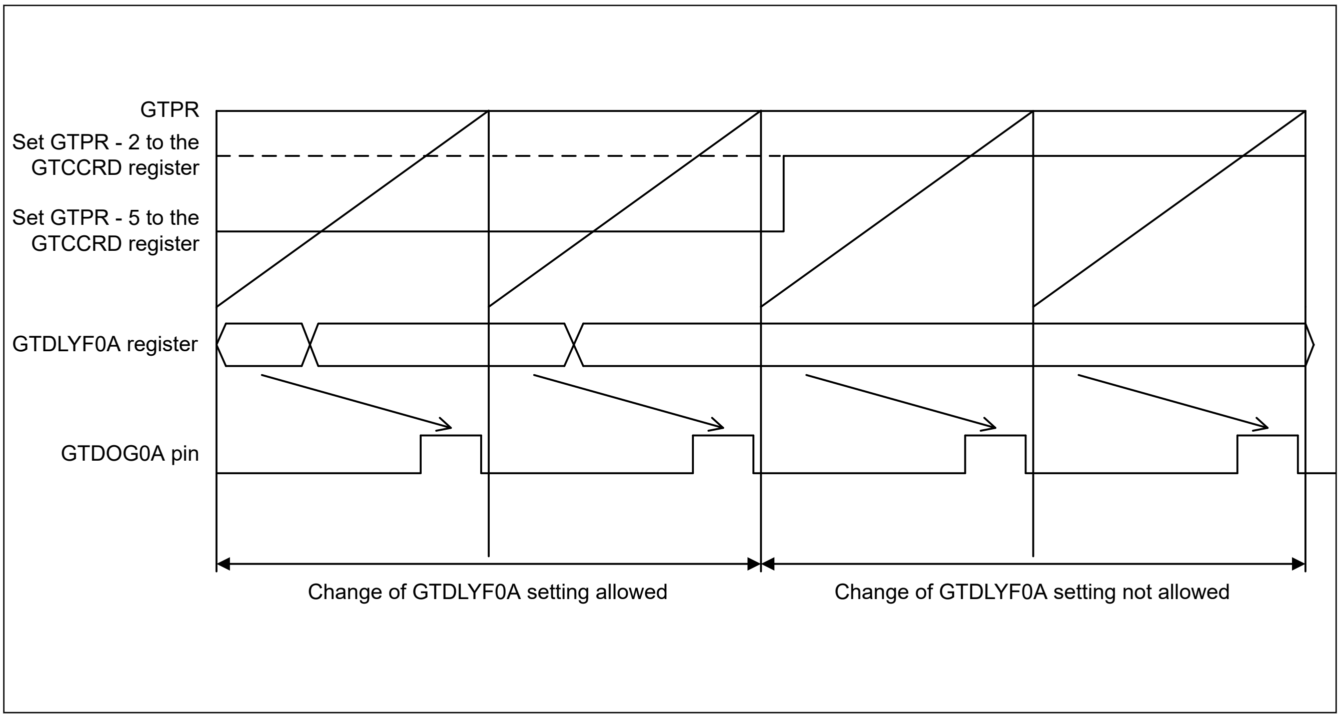



**Figure 23.5 Constraints on the timing of GTDLYF0A register settings**

Changing the values in the GTDLYFnA, GTDLYRnA, GTDLYFnB, and GTDLYRnB registers during periods where changes to settings are not allowed, might lead to faulty output waveforms such as shifts in the timing of output waveform transitions from the expected values.

### 23.4.3 Register Write Interval

When GPT core clock is GPTCLK, the writing value may not be reflected if the interval between writing to the GTDLYRnA/GTDLYFnA/GTDLYRnB/GTDLYFnA register is shorter than the following interval time. This restriction only applies to successive writes to the same register.

Write_Interval [ns] = Period_of_PCLKA [ns] × 6 + Period_of_GPTCLK [ns] × 4